University of Rajshahi

Department of Computer Science and Engineering

CSE4261 Neural Networks and Deep Learning

Assignment-10

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Problem - 1: Generating five images passing five noise vectors/matrices drawn from a normal distribution with mean 5 and variance through the Decoder of my normal Autoencoder.
As shown in Figure 1, the Auto-Encoders' output is

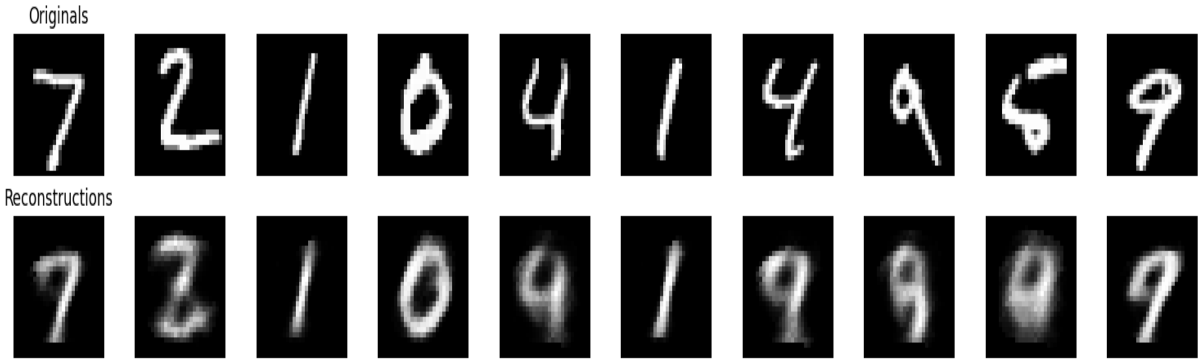

Figure 1: Output of Auto-Encoder

As shown in Figure 2,, after using this Auto-Encoder, we can generate images. The generated images using the **mean = 5** and the **std_dev = 1** are shown below.

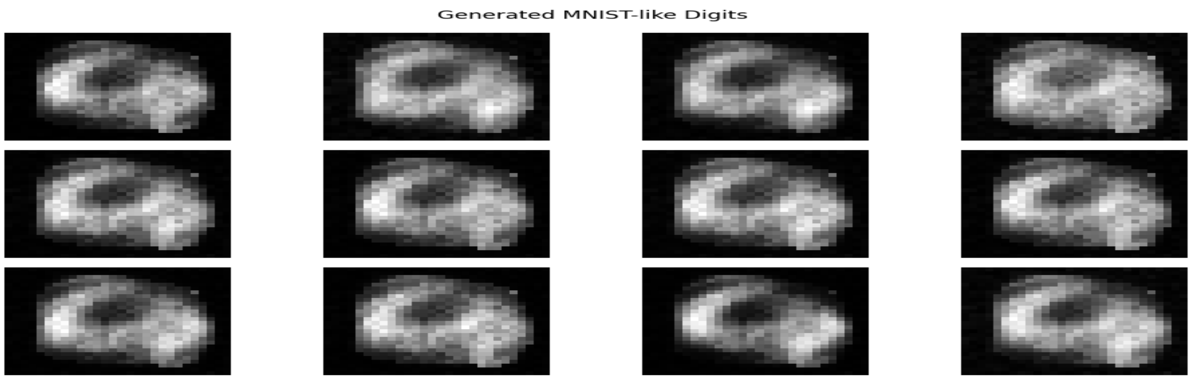

Figure 2: Output of generated images using mean = 5, std_dev = 1

As shown in Figure 3, after using this Auto-Encoder, we can generate images. The generated images using the generated function of mean and standard deviation are given below.

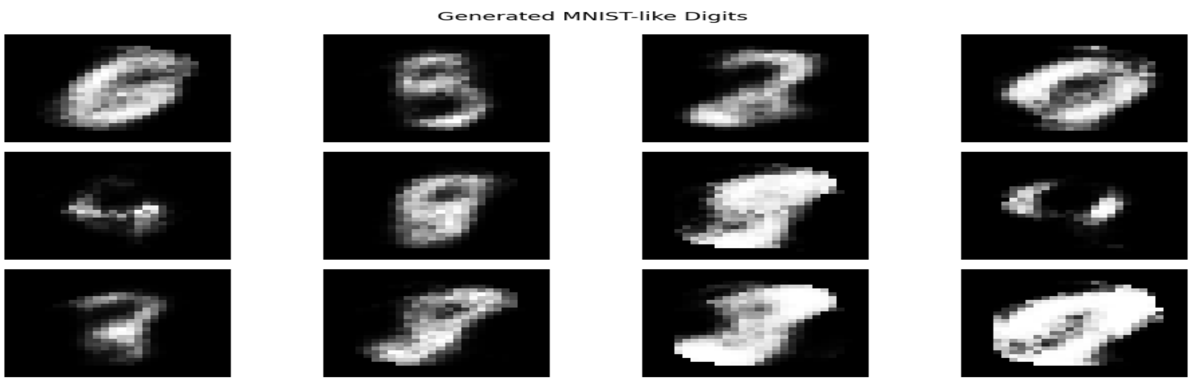

Figure 3: Output of generated images using generated mean and std_dev

Problem - 2: Generating five images passing five noise vectors/matrices drawn from a normal distribution with mean 5 and variance through the Decoder of your Denoising Autoencoder. As shown in Figure 4, the denoising auto-encoders output image.

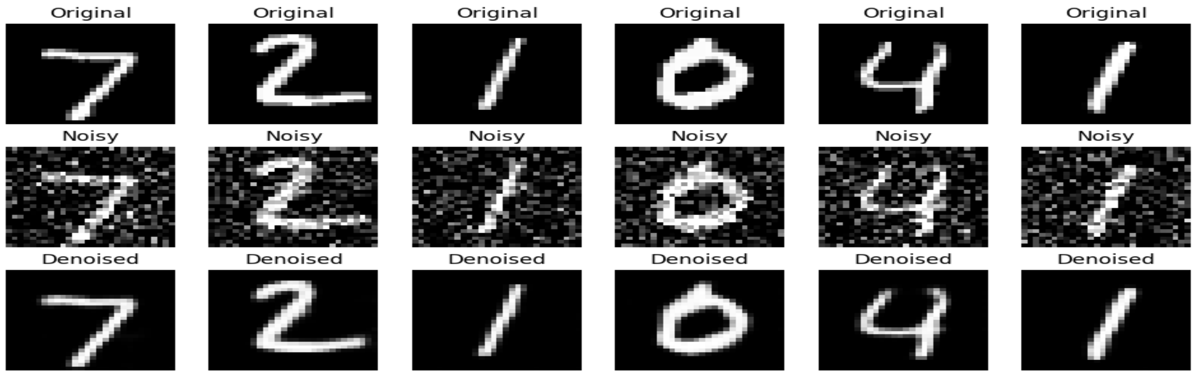

Figure 4: Output of Denoising Auto-Encoder

As shown in Figure 5, the output of denoising auto-encoders using **mean = 5** and standard deviation are find from the decoder.

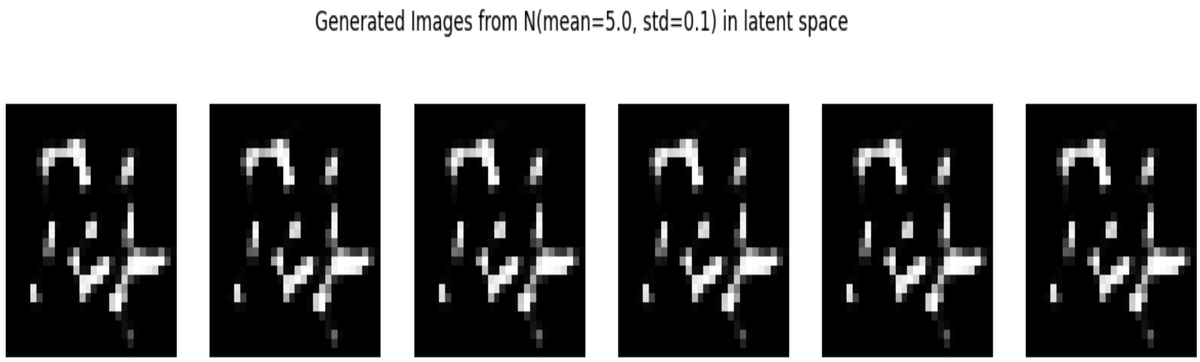

Figure 5: Output of generated images using mean = 5 and std_dev are decoders

As shown in Figure 6, the output of denoising auto-encoders using generated mean and standard deviation values.

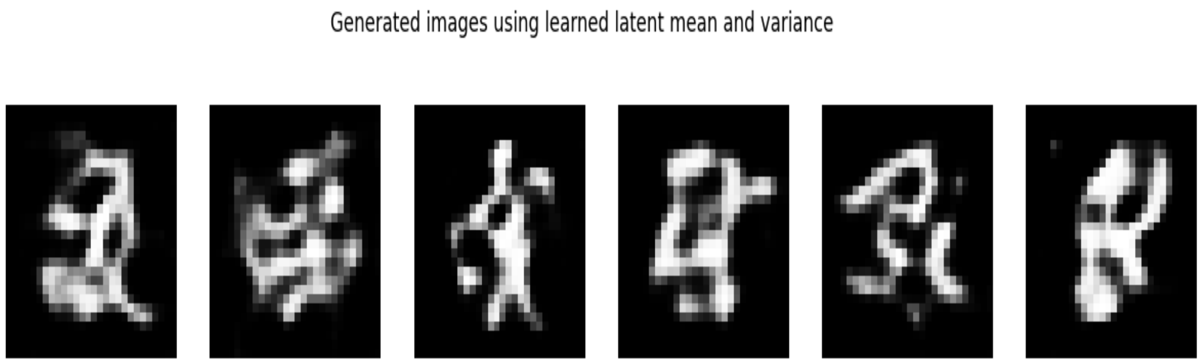

Figure 6: Output of generated images using generated mean and std_dev

Problem - 4 Training and evaluating a Variational Autoencoder.
As shown in Figure 7, the evaluated result of the Variational Autoencoder.

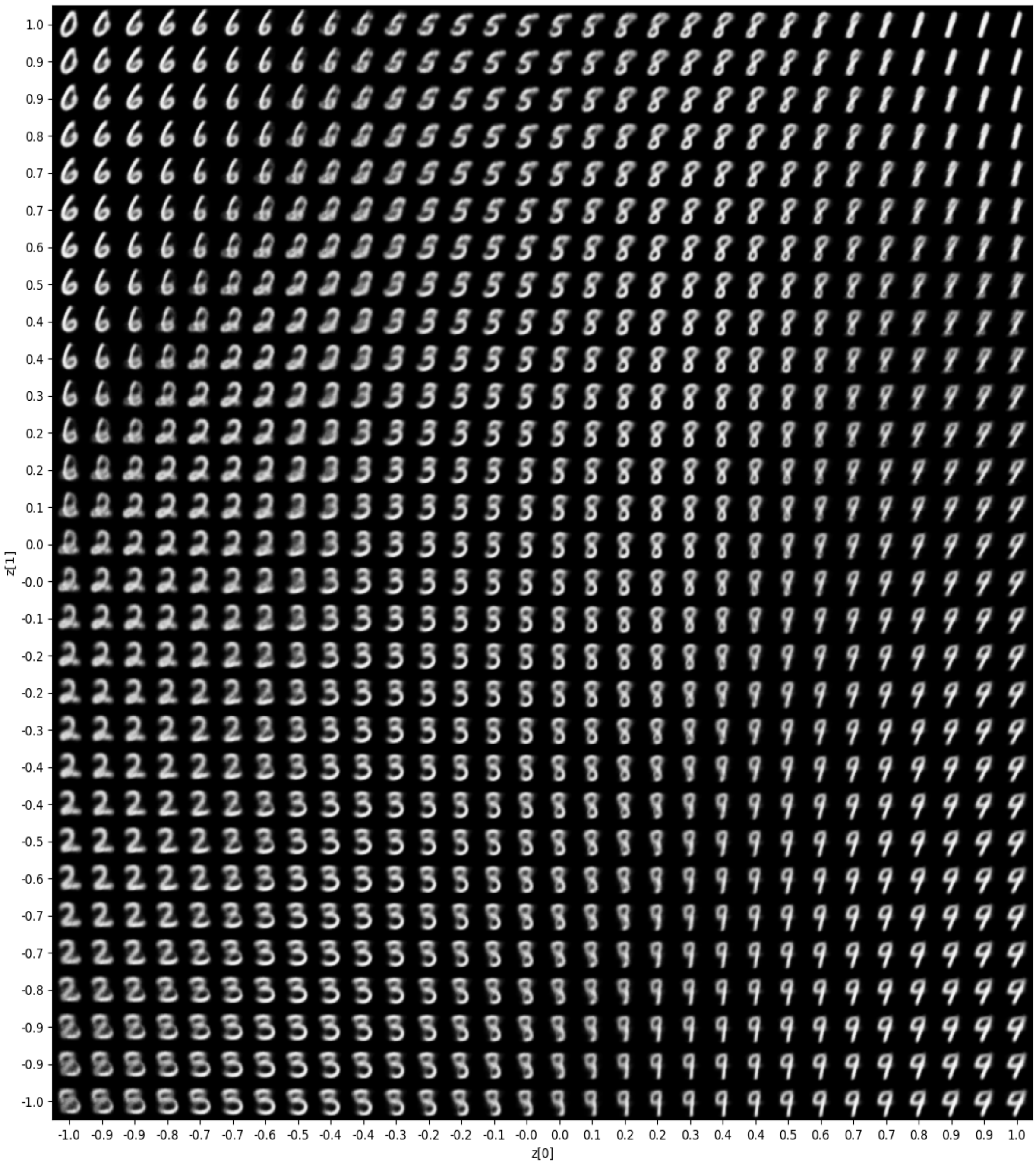

Figure 7: Output of the Variational Autoencoder

As shown in Figure 8, the digit classes clustering images.

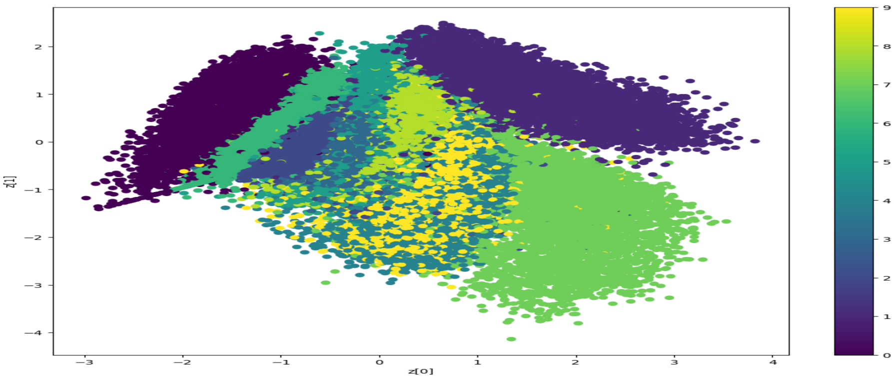

Figure 8: Clustered images

As shown in Figure 9, the generated 5 images after using the **mean = 5** and **standard deviation = 1**.

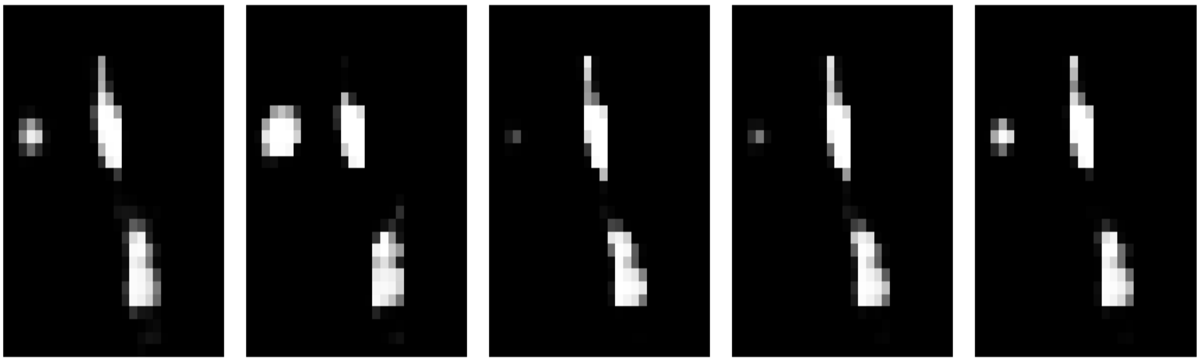

Figure 9: Generated images using mean = 5 and std dev = 1.

As shown in Figure 10, the generated 5 images after using the **mean = 0** and **standard deviation = 1**.

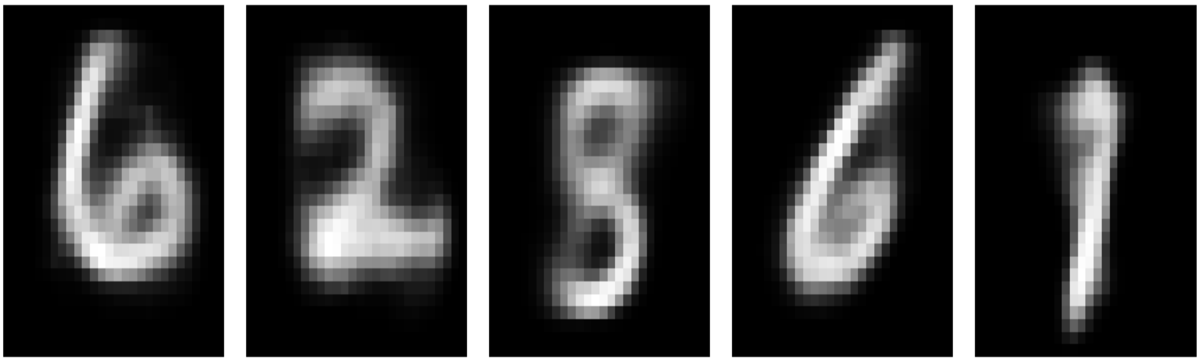

Figure 10: Generated images using mean = 0 and std dev = 1

Assignment Github Link : Assignment